

Database Technologies - Assignment 5

Use appropriate SQL functions to solve following queries.

Note: To solve below queries, use “hr” database

1. Display all the first names in lower case and the last names in upper case as the single column separated with ' - '. Provide the meaningful name to the column.

```
vishwajeet>SELECT CONCAT(LOWER(FIRST_NAME),'-',UPPER(LAST_NAME)) AS FULL_NAME FROM employees;
```

FULL_NAME
ellen-ABEL
sundar-ANDE
mozhe-ATKINSON
shelli-BAIDA
amit-BANDA
elizabeth-BATES
sarah-BELL
david-BERNSTEIN
laura-BISSOT
clara-VISHNEY
shanta-VOLLMAN
alana-WALSH
matthew-WEISS
jennifer-WHALEN
david-WILLIAMS
neena-YANG
eleni-ZLOTKEY

107 rows in set (0.052 sec)

2. Display the first word of the job title. (Hint : Use substring_index).

```
vishwajeet>SELECT SUBSTRING_INDEX(JOB_TITLE,' ',1) AS 1ST_WORD FROM jobs;
```

1ST_WORD
Public
Accounting
Administration
President
Administration
Accountant
Finance
Human
Programmer
Marketing
Marketing
Public
Purchasing
Purchasing
Sales
Sales
Shipping
Stock
Stock

19 rows in set (0.019 sec)

3. Display the First name, last name and length of first name for the employees whose last name contains character 'b' anywhere after 3rd position.

```
vishwajeet>SELECT FIRST_NAME, LAST_NAME, LENGTH(FIRST_NAME) FROM employees
-> WHERE LAST_NAME LIKE('___%b%');
+-----+-----+-----+
| FIRST_NAME | LAST_NAME | LENGTH(FIRST_NAME) |
+-----+-----+-----+
| Gerald     | Cambrault | 6                   |
| Nanette    | Cambrault | 7                   |
| Nancy      | Gruenberg | 5                   |
| Susan      | Jacobs    | 5                   |
+-----+-----+-----+
4 rows in set (0.116 sec)
```

4. Display the employee id, first name and last name of the employees whose first or last name includes a space in between.

```
vishwajeet>select EMPLOYEE_ID, FIRST_NAME, LAST_NAME FROM employees
-> WHERE FIRST_NAME LIKE('% %') OR LAST_NAME LIKE('% %');
+-----+-----+-----+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME |
+-----+-----+-----+
| 112         | Jose Manuel | Urman     |
+-----+-----+-----+
1 row in set (0.050 sec)
```

5. Display first name, salary, and round the salary to thousands.

```
vishwajeet>SELECT FIRST_NAME, SALARY, ROUND(SALARY, -3) FROM employees;
+-----+-----+-----+
| FIRST_NAME | SALARY    | ROUND(SALARY, -3) |
+-----+-----+-----+
| Steven     | 24000.00  | 24000             |
| Neena      | 17000.00  | 17000             |
| Lex        | 17000.00  | 17000             |
| Alexander  | 9000.00   | 9000              |
| Bruce      | 6000.00   | 6000              |
```

```
| Hermann    | 10000.00  | 10000             |
| Shelley    | 12008.00  | 12000             |
| William    | 8300.00   | 8000              |
+-----+-----+-----+
107 rows in set (0.040 sec)
```

6. Display employee ID and the date on which he ended his previous job in the format as mentioned. (Format: Thursday, October 27th 2011)

```
vishwajeet>SELECT EMPLOYEE_ID, END_DATE, DATE_FORMAT(END_DATE,'%W, %M %D %Y')as Job_end_date FROM job_history;
```

EMPLOYEE_ID	END_DATE	Job_end_date
101	2011-10-27	Thursday, October 27th 2011
101	2015-03-15	Sunday, March 15th 2015
102	2016-07-24	Sunday, July 24th 2016
114	2017-12-31	Sunday, December 31st 2017
122	2017-12-31	Sunday, December 31st 2017
176	2016-12-31	Saturday, December 31st 2016
176	2017-12-31	Sunday, December 31st 2017
200	2011-06-17	Friday, June 17th 2011
200	2016-12-31	Saturday, December 31st 2016
201	2017-12-19	Tuesday, December 19th 2017

10 rows in set (0.059 sec)

7. Display first name and date of first salary of the employees. (Consider 1st day of month as salary day. HINT: LAST_DAY).

```
vishwajeet>select employee_id,first_name,hire_date,DATE_ADD(LAST_DAY(hire_date),interval 1 day) as first_sal from employees;
```

employee_id	first_name	hire_date	first_sal
100	Steven	2013-06-17	2013-07-01
101	Neena	2015-09-21	2015-10-01
102	Lex	2011-01-13	2011-02-01
103	Alexander	2016-01-03	2016-02-01
104	Bruce	2017-05-21	2017-06-01
105	David	2015-06-25	2015-07-01
106	Valli	2016-02-05	2016-03-01

200	Jennifer	2013-09-17	2013-10-01
201	Michael	2014-02-17	2014-03-01
202	Pat	2015-08-17	2015-09-01
203	Susan	2012-06-07	2012-07-01
204	Hermann	2012-06-07	2012-07-01
205	Shelley	2012-06-07	2012-07-01
206	William	2012-06-07	2012-07-01

107 rows in set (0.070 sec)

8. Display first name and experience of the employees in the format of number of years, months and days completed.

```
vishwajeet>SELECT first_name, TIMESTAMPDIFF(YEAR, hire_date, CURDATE()) AS Years,
-> TIMESTAMPDIFF(MONTH, hire_date, CURDATE()) % 12 AS Months,
-> DATEDIFF(CURDATE(), hire_date) % 30 AS Days
-> FROM employees;
```

first_name	Years	Months	Days
Steven	12	8	18
Neena	10	5	2
Lex	15	1	4
Alexander	10	1	18
Bruce	8	9	24
David	10	8	0

Michael	12	0	13
Pat	10	6	7
Susan	13	8	3
Hermann	13	8	3
Shelley	13	8	3
William	13	8	3

107 rows in set (0.016 sec)

9. Display the employee id, first name and joining date of the employees who joined in the month of august of any year.

```
vishwajeet>select employee_id, first_name, hire_date from employees
-> where month(hire_date)=8;
```

employee_id	first_name	hire_date
108	Nancy	2012-08-17
109	Daniel	2012-08-16
119	Karen	2017-08-10
129	Laura	2015-08-20
134	Michael	2016-08-26
152	Peter	2015-08-20
158	Allan	2014-08-01
189	Jennifer	2015-08-13
202	Pat	2015-08-17

9 rows in set (0.083 sec)

10. Display the details of employees who joined before the year 2015.

```
vishwajeet>select * from employees
-> where hire_date<'2015-01-01';
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90
102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.00	NULL	100	90
108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	2012-08-17	FI_MGR	12000.00	NULL	101	100
109	Daniel	Faviet	DFAVIET	1.515.555.0109	2012-08-16	FI_ACCOUNT	9000.00	NULL	108	100
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN	11000.00	NULL	100	30
115	Alexander	Khoo	AKHOO	1.515.555.0115	2013-05-18	PU_CLERK	3100.00	NULL	114	30

200	Jennifer	Whalen	JWHALEN	1.515.555.0165	2013-09-17	AD_ASST	4400.00	NULL	101	10
201	Michael	Martinez	MMARTINE	1.515.555.0166	2014-02-17	MK_MAN	13000.00	NULL	100	20
203	Susan	Jacobs	SJACOBS	1.515.555.0168	2012-06-07	HR_REP	6500.00	NULL	101	40
204	Hermann	Brown	HBROWN	1.515.555.0169	2012-06-07	PR_REP	10000.00	NULL	101	70
205	Shelley	Higgins	SHIGGINS	1.515.555.0170	2012-06-07	AC_MGR	12000.00	NULL	101	110
206	William	Gietz	WGIEZT	1.515.555.0171	2012-06-07	AC_ACCOUNT	8300.00	NULL	205	110

24 rows in set (0.012 sec)

11. Display the number of days between system date and 1st January 2011.

```
vishwajeet>SELECT DATEDIFF(CURDATE(), '2011-01-01')AS DAYS;
+-----+
| DAYS |
+-----+
| 5536 |
+-----+
1 row in set (0.004 sec)
```

12. Display number of employees joined after 15th of any month.

```
vishwajeet>SELECT COUNT(EMPLOYEE_ID)AS COUNT FROM EMPLOYEES
-> WHERE DAY(HIRE_DATE)>15;
+-----+
| COUNT |
+-----+
|    57 |
+-----+
1 row in set (0.009 sec)
```

13. Display third highest salary of employees.

```
vishwajeet>SELECT DISTINCT SALARY FROM EMPLOYEES
-> ORDER BY SALARY DESC
-> LIMIT 2,1;
+-----+
| SALARY |
+-----+
| 14000.00 |
+-----+
1 row in set (0.101 sec)
```